

Project 4

Looking along the long walls, the room is now darker toward the middle than at the corners. This indicates that we have spread the lights too far apart. WE could proceed with further contour plots, but instead let's be systematic about finding the best position for the lights. In general, we can put one light at $x=d$ and the other symmetrically at $x=10-d$ for d between 0 and 5. Judging from the examples above, the darkest spots will be either at the corners or at the midpoints of the two long walls. By symmetry, the intensity will be the same at all four corners, so let's graph the intensity at one of the corners (0, 0) as a function of d .

```
>> d = linspace(0, 5);
>> I = 150./(4*pi*(d.^2 + 13)) + 150./(4*pi*((10-d).^2 + 13));
>> plot(d, I);
```

As expected, the smaller d is, the brighter the corners are. In contrast, the graph for the intensity at the midpoint (5, 0) of a long wall (again by symmetry it does not matter which of the two long walls we choose) should grow as d increases toward 5.

```
>> d = linspace(0, 5);
>> I = 300./(4*pi*((5-d).^2 + 13));
>> plot(d, I);
```

We are after the value of d for which the lower of the two numbers on the above graphs (corresponding to the darkest spot in the room) is as high as possible. We can find this value by showing both curves on one graph.

```
>> d = linspace(0, 5);
>> IC = 150./(4*pi*(d.^2 + 13)) + 150./(4*pi*((10-d).^2 + 13));
>> IM = 300./(4*pi*((5-d).^2 + 13));
>> plot(d, IC, 'b'); hold on;
>> plot(d, IM, 'r');
```

The optimal value of d is at the point of intersection, near 1.4, minimum intensity a little under 1. To get the optimum value d , we find exactly where the two curves intersect.

```
>> f = 150./(4*pi*(d.^2 + 13)) + 150./(4*pi*((10-d).^2 + 13)) - 300./(4*pi*((5-d).^2 + 13));
>> optimal = fzero(f, 1.5)
```

```
>> 1.4410
```

So, the lights should be placed about 1.44m from the short walls. For this configuration, the approximate intensity at the darkest spots on the floor is as follows.

```
>> min = 300/(4*pi*((5-1.44)^2 + 13))
>> disp(min)
```

```
>> 0.9299
```

The darkest spots in the room have intensity around 0.93, as opposed to 0.63 for a single bulb. This is an improvement of about 50%.